

Amit Kumar

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Experience

Quansight (now OpenTeams)

Staff Software Engineer

May. 2020 – Present

Remote, London

- > Core maintainer of **Nebari**, an open-source, Kubernetes-native data-science platform; automated TLS, OIDC across AWS, GCP and Azure.
- > Optimised GPU kernels in population-genetics toolkit **sgkit**, delivering a **10x speed-up** and cutting cohort-analysis runtimes from hours to minutes (CUDA, Dask, RAPIDS).
- > Led backend work on various client services (FastAPI, Flask, Django)
- > Built a **JupyterHub extension** that launches sandboxed Streamlit, Gradio and Panel apps on Kubernetes cluster and local.
- > Contributed to numerous Data Engineering, Data Science Infrastructure, Backend, Finance client problems using tools in PyData Ecosystem like Dask, Numba, RAPIDS, etc.

Cirun.io

Founder & Principal Engineer

June 2020 – Present

Remote

- > Architected a **multi-cloud CI/CD platform** that transparently schedules GitHub Actions on **6 providers** (AWS, GCP, Azure, OCI, DigitalOcean, OpenStack) using Celery for task queuing
- > Serves over **100 000+ API requests daily** through **Cloudflare Workers**
- > Built a real-time observability system via Beszel and led a 24x7 on-call rotation, sustaining **99.9 %** platform uptime.
- > Tech Stack: Python (FastAPI, Celery), Postgres, Redis, Terraform, GitHub Actions API, Cloudflare Workers.

Contract Software Engineer

Software Engineer

June. 2018 – Mar. 2020

London, UK

- > Worked as contract Software Engineer for Aire Labs (Fintech), Turner & Townsend, Cambridge Medical Robotics
- > Designed and implemented client data sharing service and infra for credit assessment system.
- > Wrote micro service for client assessment using AWS Lambda.
- > Designed and implemented numerous API's to make construction data accessible to the business.
- > Lead the standardization of design and implementation of Data Pipelines using Airflow.
- > Designed and implemented a System Testing Infrastructure of a component of the Surgical Robot.
- > Responsible for Code Review/Mentoring and reporting, diagnosing and fixing software issues in the Robot.
- > Lead the standardization of design and implementation of Data Pipelines using Airflow.

The Telegraph

Software Engineer, Data Platform

Feb. 2018 – May. 2018

London, UK

- > Designed and implemented numerous ETL pipelines, improved deployment Infrastructure with Jenkins.
- > Improved standardization of Python Projects and lead the migration of Python2 to Python3 for ETL pipelines.

Powervault

Software Engineer, Backend

Oct. 2017 – Feb. 2018

London, UK

- › Integrated external services to control IoT devices and improved usability of client facing dashboards.
- › Dockerized the backend stack and implemented CI/CD on GitLab for 1-click test and deploy.

Zomato

Software Development Engineer

Oct. 2016 – Oct. 2017

Gurgaon, India

- › Improved user credibility system and rewrote a rule based spam detection system.
- › Created numerous internal dashboards for visualization.
- › Wrote data pipeline for image quality prediction and a cuisine predictor using word embeddings.

Google Summer of Code 2015-2020

Python Software Foundation, SymPy

Summer 2015

Remote

- › Improved Mathematical robustness of new equation solvers module in SymPy.
- › Implemented Complex Sets: Representing infinite solutions in the argand plane.
- › Linear system solver, differential calculus methods.
- › Mentored students in Google Summer of Code 2016-2020 under SymPy.

Education

Delhi Technological University

Bachelor of Technology in Mathematics and Computing Engineering

2012 – 2016

New Delhi, India

Presentations

Talks and Presentations

Meetups and Conferences

2016 - 2022

- › I have spoken at numerous meetups and conferences, all of them can be seen here: iamit.in/talks

Projects

cirun.io A service to create Github Actions CI runners on AWS, GCP, Azure, Oracle, OpenStack, DigitalOcean.

codecinema.io A tool to create and share beautiful images and videos of code.

ufo.camera Fetches random images of UFO from National UFO Reporting Center.

gitspace.io Calculates the storage space occupied by user's public GitHub repositories.

Skills

Languages Python, JavaScript, HTML/CSS, Bash, Java (Familiar), C, C++ (Familiar)

Tools, Technology & Platform Flask, Django, FastAPI, Docker, K8s, REST API, Git, TDD, Terraform, React, SQL, AWS, GCP, Azure, OpenStack, Nginx, Traefik, Dask, JupyterHub, Pytest, etc

Honors & Awards

Exceptional Talent in Technology (Endorsement) by Tech City UK (Government of UK)

Financial Aid Award Presentation on SymPy at SciPy 2016

SemiFinal HackerRank WorldCup coding competition (Team size: 2)

Open Source Contributions

- Nebari (Core Developer), SymPy (Core Developer), Panel, sgkit, pyflyby, jhub-apps, SymEngine, SymPy-Gamma, Papyri, Dask Gateway, NumPy, Flask-Caching, CPython, pandas-gbp, facebook-prophet, pydsa